

# Jeroen Van Goey

Staff Research Engineer in BioAI building generative models for biology

📍 Cape Town, South Africa    ✉ jeroen.vangoey@gmail.com    ☎ 072 458 2796    🌐 jeroen.vangoey.be    🌐 jeroenvangoey  
 🧑‍🔬 BioGeek    🎓 Google Scholar    🆔 0000-0003-4480-5567

## Summary

Machine learning research engineer working at the intersection of AI and science, bridging scientific ML, biological domain knowledge, production-quality software, and team leadership. I build generative models and take them from prototype to deployment, with a focus on *de novo* peptide sequencing.

8 publications, including in *Nature Machine Intelligence*.

Currently looking for Staff/Principal Research Engineer roles in AI for science. Based in Cape Town, open to remote-first roles now and hybrid/on-site roles in Europe from 2027.

## Experience

**InstaDeep**, Staff Research Engineer in BioAI

Cape Town, South Africa

Aug 2022 – present

3 years 11 months

- **Tech lead and hiring manager** for two cross-disciplinary **ML research teams** of about 5 to 8 people: *de novo* peptide sequencing (with the Technical University of Denmark) and signal-peptide design for secretion efficiency (with BioNTech).
- Delivered a **client collaboration with Syngenta**, applying **genomic language models** (AgroNT, a Nucleotide Transformer trained on ~10.5 million genomic sequences spanning trillions of base pairs across 48 plant species) to accelerate crop trait research.
- Set **research direction** and take models **from research to production** (InstaNovo is available to commercial customers on **DeepChain**), balancing scientific rigour with reliable engineering.
- **Lead the BioAI department** of InstaDeep's Cape Town office and serve as the office's **site manager** (an office of about 25 people); hire and grow the BioAI teams across the Cape Town and Kigali offices.
- Directed the team's **ML engineering foundations**: scalable, Transformer-based libraries (Python / PyTorch) for large-scale training on InstaDeep's Kyber (~500 PFLOPs) and EuroHPC's MareNostrum 5 (260 PFLOPs) supercomputers, and the cloud (AWS, GCP).

**Barco**, Senior Software Development Engineer, Machine Learning

Kortrijk, Belgium

Feb 2020 – Aug 2022

2 years 7 months

- Built a TensorFlow Extended production pipeline (orchestrated with Apache Airflow) training deep-learning models on multispectral images to detect and classify melanoma skin cancers for Demetra, Barco's dermatology imaging device.
- Engineered the pipeline with **end-to-end data and model lineage tracking** to deliver the traceability and reproducibility that **regulated medical industries (FDA approval)** require.
- Worked across the research, cloud-backend and mobile/web frontend teams.

**Bayer Crop Science, then BASF**, Bioinformatics Researcher, Manager of the Python & R Platforms

Zwijnaarde, Belgium

Feb 2018 – Jan 2020

2 years

- Owned the **Python/R data-analysis platform** used by **480 researchers and data scientists**: technical support, proactive monitoring and root-cause analysis, **training** (NumPy, pandas, BioPython) and **mentoring** across the company's internal research community.
- The Zwijnaarde site was divested from Bayer Crop Science to BASF as part of Bayer's acquisition of Monsanto; I continued in the same role across the transition.

**Applied Maths (acquired by bioMérieux)**, Bioinformatics Software Developer

Sint-Martens-Latem, Belgium

Sept 2011 – Jan 2018

6 years 5 months

- Worked on BioNumerics, a bioinformatics suite for integrated analysis of biological data, writing **custom Python scripts and extensions** for clients in the clinical, academic and government sectors.

## Selected publications

**InstaNovo enables diffusion-powered *de novo* peptide sequencing in large-scale proteomics experiments**

Nature Machine Intelligence  
2025

K. Eloff, K. Kalogeropoulos, ..., **J. Van Goey**, ..., T. P. Jenkins

Trained on ~63 million spectra; top-ranked in an independent benchmark of 17 *de novo* tools across 83 datasets; Altmetric attention score 100+ and 120+ GitHub stars.

[10.1038/s42256-025-01019-5](https://doi.org/10.1038/s42256-025-01019-5)

**InstaNovo-P: a *de novo* peptide sequencing model for phosphoproteomics**

bioRxiv

J. Lauridsen, P. Ramasamy, ..., **J. Van Goey**, ..., K. Kalogeropoulos

2025

[10.1101/2025.05.14.654049](https://doi.org/10.1101/2025.05.14.654049)

### De novo peptide sequencing rescoring and FDR estimation with Winnov

A. Mabona, J. Daniel, ..., **J. Van Goey**, K. Kalogeropoulos

[10.48550/arXiv.2509.24952](https://doi.org/10.48550/arXiv.2509.24952)

arXiv

2025

### A living proteomics benchmark for comprehensive evaluation of deep learning-based de novo peptide sequencing tools

Registered report (preprint)

2026

M. Pominova, **J. Van Goey**, C. Adams, ..., W. Bittremieux

[10.6084/m9.figshare.31680883.v1](https://doi.org/10.6084/m9.figshare.31680883.v1)

### Generalizable direct protein sequencing with InstaNexus

Mol. Cell. Proteomics

2026

M. Reverenna, M. Wennekers Nielsen, ..., **J. Van Goey**, ..., K. Kalogeropoulos

[10.1016/j.mcpro.2026.101547](https://doi.org/10.1016/j.mcpro.2026.101547)

### A Framework for Database Search with AI Models in Mass Spectrometry-Based Proteomics

Journal of Proteome Research

2026

K. Kalogeropoulos, **J. Van Goey**, T. P. Jenkins, K. M. Eloff

[10.1021/acs.jproteome.5c01153](https://doi.org/10.1021/acs.jproteome.5c01153)

## Education

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**HOWEST Hogeschool West-Vlaanderen**, Microdegree Machine Learning & Deep Learning

Kortrijk, Belgium

2018 – 2019

**Katholieke Universiteit Leuven**, Bioinformatics (partial credits obtained)

Leuven, Belgium

2010 – 2012

**University of Antwerp**, M.Sc. Biology

Antwerp, Belgium

1997 – 2004

**Universitat de Barcelona**, Erasmus interuniversity exchange

Barcelona, Spain

2002

**Møglestu videregående skole**, AFS intercultural exchange program

Lillesand, Norway

1996 – 1997

## Skills

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**Machine learning:** Transformers, diffusion models, large language models (LLMs), supervised & self-supervised learning

**Frameworks:** PyTorch, TensorFlow / TFX, NumPy, pandas

**MLOps:** Kubernetes-based ML platforms (**Alchor**), Docker, CI/CD, workflow orchestration (Airflow, Dagster), experiment tracking (MLflow, Neptune, TensorBoard)

**Cloud:** Buckets, VMs, Cloud Run (AWS, GCP)

**Scale:** Distributed GPU training, HPC clusters (InstaDeep's Kyber and EuroHPC's MareNostrum 5)

**Bioinformatics:** BioPython, BLAST, ClustalW, Snakemake, BioNumerics

**Languages:** Python (primary), R

## Projects

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### Awesome De Novo Peptide Sequencing

A comprehensive, interactive map of the *de novo* peptide sequencing field: algorithms, post-processors, downstream applications, and adjacent tools, deep-learning and classical alike.

- Open source: [github.com/BioGeek/awesome\\_de\\_novo\\_peptide\\_sequencing](https://github.com/BioGeek/awesome_de_novo_peptide_sequencing)

## Teaching & mentoring

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### Machine Learning for Biology practical

Deep Learning Indaba, Ghana

Mentored the hands-on practical alongside InstaDeep and Google DeepMind colleagues.

2023

### Scientific Python training & mentoring

BASF and Bayer Crop Science,

Trained and mentored researchers and data scientists in scientific Python (NumPy, pandas, BioPython)

Belgium

as manager of the Python/R analysis platform.

2018 – 2020

## Hackathons (co-created, organized and judged)

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**2025:** Machine Learning Interatomic Potentials Hackathon, IndabaX (Stellenbosch University, South Africa)

**2024:** Snakes and Sequences: Senegalese Serpent Venom Sequencing Hackathon, Deep Learning Indaba (Senegal)

**2024:** Desert Locust Breeding Ground Prediction Hackathon, IndabaX (Wits, Johannesburg)

**2023:** Unveiling Cassava's Secrets, Deep Learning Indaba (Ghana)